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Dissolvable Plugged Nozzles Designed to Maintain Integrity at 220°F, 3500 psi Differential for (a) Up-To 20-days in RDFNAF (b) 15-hrs. in 10.6 ppg Brine (c) 8-hrs. in Acidic Breaker

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Abstract

ADNOC-offshore with their partners conceived and developed the concept of Dissolvable-Plugged-Nozzles (DPNs'), developing an inflow-control-device (ICD) with dissolvable elastomeric-plug to be installed on limited-entry perforated-liners in their extended-reach and deviated (ERD) injection-wells (Bachar *et al.* 2004, Beaman *et al.* 2016, Clews *et al.* 2018, Jackson *et al.* 2017, Kiyouni *et al.* 2019, Mayer *et al.* 2014, and Sau *et al.* 2019). Benefits of DPNs' encompass significant savings of the order of several- million-dollars for the client due to one or more of the benefits (a) Not having to pick-up a dedicated displacement work-string, thus significant time-savings of up to 5-days (b) Increase probability of running liner to TD through the ability to wash to bottom (c) Less fluid losses to the reservoir formation (d) Effective wellbore cleaning, among others (Roy *et al.* 2021, Roy *et al.* 2020, and Shamshy *et al.* 2023). With extended reach deviated island wells now reaching > 50,000 ft., often having a lateral length of > 30,000 ft., substantial resistance is encountered during deploying completions string to total depth (TD), necessitating jarring of the LEL assembly while running in hole (RIH), resulting in days of delay. Use of acidic breaker to reduce differential sticking necessitates qualifying a DPN wherein it retains mechanical integrity in breaker fluids for a minimum of 8 hours.

This article encompasses underlying design methodology and includes verification and validation (V&V) testing results of multiple DPNs' potted with low ROC partially vitrified high entropy alloy (HEA) plugs when exposed to 10.6 ppg brine and breaker. Objective of this qualification program was to identify the Breaker-Resist (BR)-DPN potted with the "specific plug chemistry", which reliably maintains mechanical integrity for extended time durations, 15 hours in buffered 10.6 ppg brine followed by 8-hours (+FOS) or more in 50/50 10.6 ppg brine/breaker.

This was achieved with our HEA5 formulation. Validation success led to a greenlight by operator and approval for manufacturing needed number of BR-DPNs' for field introduction, led by our commercialization partner, Baker Hughes. Our rigorous testing program has demonstrated promise of

successfully establishing, including concept of a BR-DPN offering, patent pending "Dissolvable Plugged Nozzle Having Single or Multi-Layer Breaker-Resist Core", 63/658,864 dated 06-11-24 and granted Patent 11/719074 dated 08-08-23.

In this study all evaluations have been conducted with un-inhibited aggressive 10.6 ppg brine and Breaker I. In the interest of time, per operator's advice we focused on testing and repeatably determined breakthrough times in available fluids. HEA5 formulation bought us a direct breakthrough time of 50 ± 10 hours in uninhibited 10.6 ppg brine. Operational tests in the aggressive 50/50 10.6 ppg brine/Breaker I bought >37 hours till breakthrough after 15 hours of exposure in uninhibited 10.6 ppg brine. They lasted 15+ hours in an earlier inhibited batch of 50/50 10.6 ppg brine/Breaker I after 15 hours of exposure in the old buffered 10.6 ppg brine. The rationale for HEA5 to maintain mechanical integrity (>37 hrs. vs. reduced time of 15+ hrs.) is unknown. It may be possibly (1) due to breakdown of the inhibited breaker over time (2) because of the added inhibitors it might have coated pressure vessel surfaces, preventing reaction between the fluid and vessel metallurgy, preserving activity of the fluid. The Breaker I with hydronium ions (H_3O^+) preserved may have resulted in accelerated degradation of the installed DPN. This is representative and the most likely case, wherein an active, flowing inhibited breaker will interact with BR-DPNs' in the wellbore. Our HEA5 (and all developed formulations) have repeatedly survived 15 hrs. in 10.6 ppg brine and $>8+$ factor of safety (FOS) hours in uninhibited breaker or old inhibited counterpart.

Introduction

Venturi Dissolvable Plugged Nozzle (DPN) aka Salinity Insensitive Metallic-Glass Barrier Assembly (SIMBA) BL-F were qualified in operator specified brines for up to 48-hours under a 5000-psi differential pressure (DP). Tests were conducted to determine the breakthrough time for SIMBA BL-F in 10.6 ppg buffered brine followed by breaker at 220 °F, subjected to a DP of 3500 psi (ID to OD). Initial Exposure was in 10.6 ppg low pH brine for 15-hrs. Subsequent exposure was in fluid composed of equal parts (50% 10.6 ppg brine, 50% breaker), until communication through Venturi DPN i.e. breakthrough.

The first phase of the test revealed that the SIMBA BL-F reacted to given 10.6 ppg brine, however, was mostly unaffected for the entire 15 hours and held pressure differential across the DPN. In the second phase of testing, the SIMBA BL-F DPN (a) breached (broke through) in 5 hrs. 32 mins in a breaker fluid composed of equal parts (50% brine, 50% breaker) of 10.6 ppg brine and Breaker I, (b) breached (broke through) in 11 hrs. and 1 min in Breaker II.

SIMBA BL-F is designed for 10.6 ppg or other brines, and not for a fluid (as such a breaker) with a pH 2 or lower at operating conditions. Conducting this qualification does establish a calibration point, the duration to which SIMBA BL-F will survive in breaker(s) post exposure to brine(s); also, these tests will provide the platform on which breaker resist and multilayered formulations* (patents pending) can be designed.

Operating envelope of BR-DPN specifies that a minimum of 8-hours is needed for the plugged nozzles to survive in breaker. SIMBA BL-F fell short of the 8-hour benchmark when exposed to breaker I. To overcome the short-fall of SIMBA BL-F in low pH breakers, Breaker-Resist formulations were engineered.

From Formulating Blends to Manufacturing Breaker Resist DPNs' to Qualification

Degradable alloys plugging the DPN cavity are designed to a certain rate of corrosion (ROC) [Target ROC – 0] aka degradation or mass loss in aqueous brines, which are (a) 8% NaCl = 1.05 SG, (b) 21% NaCl + 9% NaBr = 1.25 SG, (c) 1% NaCl + 43% NaBr = 1.49 SG (per client DPN design specifications). A range of formulations were designed which have predicted lower ROCs' in target environments. [Figure 1](#) shows our design methodology to vary reactive complexes in HEA DPN plug formulation, wherein engineered material results in unique breakthrough times in specific environmental conditions: Brine, breaker, temperature, pressure, among others ([Roy et al., 2021, 2025](#)).



Figure 1—Design methodology to vary reactive complexes in HEA DPN plug formulation.

As presented elsewhere in our earlier publications, (Roy *et al.* 202, 2025), we have designed our HEA multi-component degradable BR-DPN plug alloy with a novel approach of combining grain boundary engineered (GBE) nanocrystalline alloys with controlled volume fraction of partially or fully vitrified "Bulk Metallic Glasses". Herein we have created this unique class of novel nanocomposite which manifest ultra-high strength, tailored reactivity aka degradation in select engineered to order fluids, however, exhibit unique properties, for example tailored de-passivation in target environments at high pressures and temperatures. The partially vitrified, high entropy alloy degradable plug of DPN has certain, "Active Complexes" (AC). Figure 2 (a) shows TEM photomicrograph with Selected Area Diffraction (SAD) inserts detailing the interface between crystalline and a partially vitrified region. High resolution image outlining features, wherein a mix of crystalline and amorphous regions are evident. SAD patterns of the amorphous region reveal the characteristic halo ring pattern and diffraction pattern of the polycrystalline region features consecutive, continuous rings. The distance of these rings from the origin can be used for the determination of interplanar spacings. These novel glassy alloys allow the degradation control needed to manufacture DPN and other embodiments where precision is paramount. These alloys form the plug-core of our DPN plug.

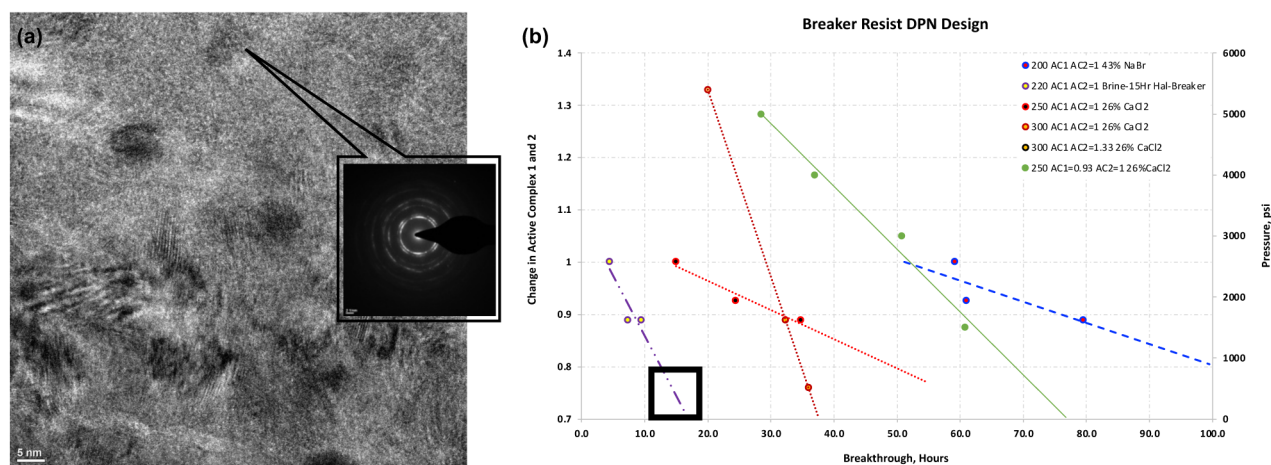


Figure 2—(a) TEM photomicrograph with Selected Area Diffraction (SAD) inserts detailing the interface between crystalline and a partially vitrified region. High resolution image outlining features, wherein a mix of crystalline and amorphous regions are evident. SAD patterns of the amorphous region reveal the characteristic halo ring pattern and diffraction pattern of the polycrystalline region features consecutive, continuous rings. (b) Plot of breakthrough times of DPNs' wherein select active complex ingredients (in this case 1 and 2) of partially vitrified HEA alloys are varied to affect its reactivity.

Here we present select data stemming from experimentation conducted to resolve sensitivity of designed formulations to degradation in target environments (brine, breaker, pressure and temperature) when we vary one or more complexes (AC 1 and 2) of HEA plug blend. It is evident from experiments conducted at 250 °F (see Figure 2 (b)): With 7% lower AC1 (from baseline, keeping AC2 constant), pressure increase resulted in a distinct reduction of breakthrough time; Lowering of exposure temperature extended breakthrough times; Addition of AC2 (33%) also resulted in lowering the breakthrough time significantly. It is hypothesized that lowering of AC1 between 20 and 30% while simultaneously tailoring other complexes will reliably get

us a breakthrough time of 10-15 hours in client supplied low pH breaker (green box), post brine exposure. Adopting this approach, a series of formulations were developed.

Samples were manufactured through powder metallurgy route, wherein formulations were unique high entropy alloys (HEA). When potted into a DPN cavity, they fully or partially vitrify, manifesting a tailored degradation rate in a given environment. Table 1, presented below, captures formulations prepared for conducting verification and validation (V&V) tests to converge on a reliable offering for initial field introduction surviving 15-Hrs exposure in buffered 10.6 ppg brine followed by 8-Hrs (+FOS) or more in 50/50 10.6 ppg brine/breaker.

Nested autoclave test set up is detailed in Figure 3. Formulations were consolidated into (green compacts) preforms. Spring-backs during consolidation and co-efficient of thermal expansion during heating cycles of preforms were diligently measured to accommodate them in vitrification tooling and is presented in Table 2.

Table 1 (Left)

Formulations	Brine (A)	Breaker (B)	Sequence
HEA1			Calibration Point
HEA2			Starting Point
HEA3			Test in Brine 15 hours hold 50:50 Brine-Breaker Till Breakthrough Based on Results "Repeat"
HEA5			
HEA6			
HEA7			
HEA8			
HEA14			
HEA15			
HEA16			
HEA17			

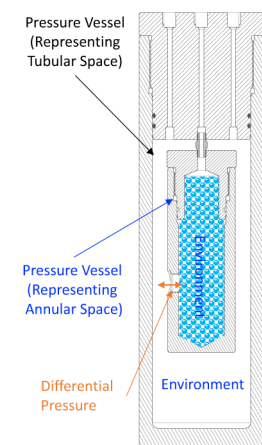


Figure 3—(Right) Nested autoclave test set up to conduct V&V tests to converge on field introduction formulation.

Table 2

Sample #:	Consolidation (% of Theory)	OD (in)	Height (in)	Fill Ratio	Spring back (%)
4839-D01	86.97	1.362	0.492	1.78	0.89
4839-D02	88.46	1.362	0.5415	1.79	0.89
4839-D03	90.71	1.362	0.566	1.83	0.89
4839-D04	92.04	1.362	0.558	1.87	0.89
4839-D05	92.65	1.362	0.554	1.89	0.89
4839-D06	92.38	1.3625	0.556	1.92	0.93
4839-D07	84.57	1.362	0.547	1.74	0.89

Table 2 also captures assessment of the spring back during consolidation of preforms of a representative blend. 100 preforms were made as detailed in Table 1, arranged systematically on individual trays, introduced in an oven and heated to the right processing temperature and potted into DPN blanks – see Figure 4 (a) through (d).

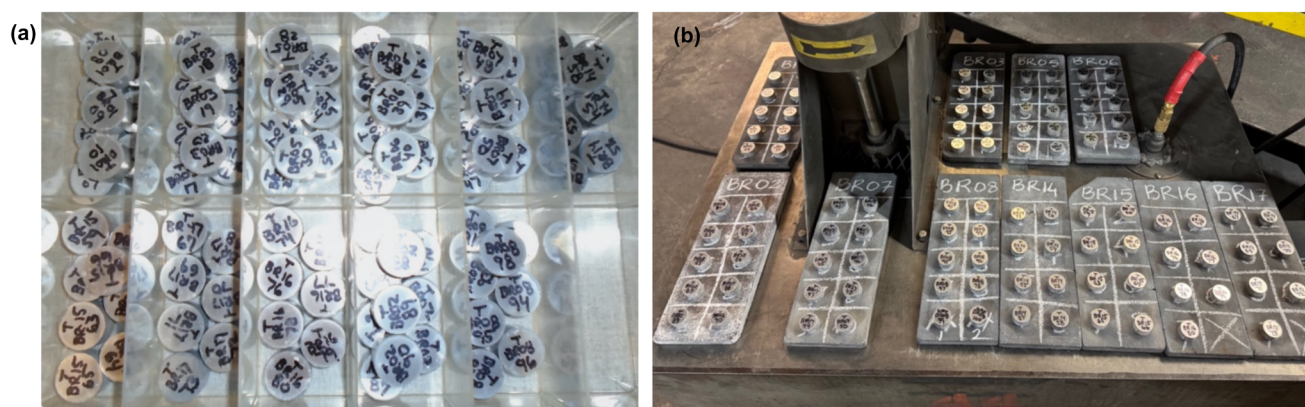


Figure 4(a)—100 preforms were consolidated as detailed in Table 1 for introduction into DPN blanks (potting). (b) Preforms systematically arranged on individual trays before introduction into oven.



Figure 4(c)—100 preforms introduced into high temperature oven in inert (flowing Argon environment) and heated to the right processing temperature (d) Potted into DPN blanks.

Potted blanks were machined finished into different BR-Blends (as detailed in Table 1) for V&V.

Test Objectives

The purpose of this screening test was to determine the breakthrough time for a Dissolvable Plugged Nozzle (DPN) with a low ROC BMG plug, wherein the plug is designed from Salinity Insensitive Metallic-Glass, qualified in RDFNAF for up to 20 days at different pressures – temperatures. The current DPNs' were tested at 220 °F, subjected to a differential pressure of 3500 psi (ID to OD) during exposure in the following environments: (a) Initial Exposure – In 10.6 ppg low pH (buffered) brine for 15 hours; (b) Second Exposure in a fluid composed of equal parts (50% brine, 50% breaker) of 10.6 ppg brine and breaker. The qualification criteria of this screening test were to survive 15 hours in 10.6 ppg brine and a minimum of 8 hours + (0.8 hours) 48 min, a 10% factor of safety (FOS). Measured was the time for DPN to communicate i.e. breakthrough.

This test is a part of the Validation and Verification plan to screen and establish that DPNs' with engineered to order BMG plugs of tailored low ROC in breakers can be easily designed to survive a specific duration in acidic and aggressive breaker fluids for immediate and upcoming field introduction.

Survival of the DPN with a low ROC BMG plug in brine followed by breaker (even up to several hours of exposure in breaker) establishes the superiority of offered DPN when compared to other commercial offerings.

Measured Parameters

The measured parameters, documented for these tests include:

- Volume of fluids
- Length of time of exposure
- Temperature
- Pressure
- Composition of mixtures used in tests
- pH of the solutions before and after tests

Materials and Methods

Hardware used for Verification & Validation (V&V)

Table 3—BR-DPNs' Tested

P/N	Rev	Description	U of M	Qty
E1000390P03	AC	DPN with Low ROC BMG Plug	Each	40+

Table 4—Test Fixture, Parts and Supporting Equipment

Description	S/N
High Pressure Pump	DA-GD-20
Air Compressor – California Air Tool	DA-CAT-20
External Pressure Vessel - Head, 5.5" OD, 4.24" ID, 11.5" Inside Depth – qty. 2	DA-PV1-925-20
O-Ring Size 2-242 for External Vessel, qty. 4	2-242
Internal Pressure Vessel with Head, 3" OD, 2" ID, 5" Inside Depth – qty. 2	DA-I-S-PV1-LAS-21
O-Ring Size 2-135 for Internal Vessel, qty. 2	2-135
Digital Pressure Gauges, Additel 680 – qty.6.	DA-AD680-21
Pressure Relief Valve – Spirstar - – qty. 4	SS-10RV-FA
Valve – Spirstar, qty. 2	SS-NVNF-15K
Hose, Swivel, 8 ft. long – Spirstar, qty. 4	6/2WL
T-Connection – Spirstar	HTNF4
Anti-Seize Lubricant – Permatex	09128
Accumulator, 5,000 psi rated, charged to 3,000 psi – Parker – qty. 2	A3N0058C1K
Accumulator, 3,000 psi rated, charged to 350 psi – Parker – qty. 2	A3N0116D1K
Torque Wrench	DA-TW-N1-21
Temperature Controller, Inkbird – qty. 2	ITC-106VH
Thermocouples – qty.4	TJ36-CAIN-116U-12
Temperate Data Logger – 4 channel capability	DA-TL-23

Testing Environments & Conditions

Environment 1: Conducted test in 10.6 ppg brine, for 15 hours.

Environment 1: Conducted test further exposing DPN with low ROC BMG plug exposed in Env. 1 in a fluid composed of equal parts (50% brine, 50% breaker) of 10.6 ppg brine and breaker. Test to be run until communication through DPN with low ROC

BMG plug i.e. breakthrough. Wherein, test conditions for the tests presented above are: 220 °F, 3500 psi ΔP (ID to OD).

Environment 1

1. External pressure vessel was filled with 10.6 ppg brine. Brine level in the pressure vessel was not exceed a height of 10" from the top of the autoclave. Fluid was heated to 220 °F.
2. DPN with low ROC BMG plug was installed on the internal pressure vessel using approximately 80 ft-lbs. of torque.
3. Internal pressure vessel was filled (topped) with 10.6 ppg brine.
4. Internal pressure vessel was attached to the bottom of the head of external pressure vessel.
5. The combo-heads attached together and communicating (internal or external) was installed on the external pressure vessel.
6. Gauge, pressure relieve valve, and adaptors were attached to the central port of the external head.
7. Using a HPHT swivel hose, an accumulator was attached to the central port of the external pressure vessel head (accumulator is pre-energized to 2500 psi).
8. Thermocouple was installed on an outer port on the external pressure vessel head. The thermocouple was connected to a temperature controller, where a stable temperature of 220 °F temperature was maintained.
9. A second HPHT swivel hose, a drainage hose, was attached to an outer port of the external pressure vessel head. A pressure gauge, a pressure relieve valve and a check valve assembly were installed at the end of this drainage hose.
10. Pressure data was recorded with a digital pressure gauge and temperature data with a temperature logger.
11. The inner vessel was pressurized to 3,500 psi minimum pressure after the system temperature reached 220 °F and was stabilized.
12. With pressure stabilizing, a minimum differential pressure across the nozzle from tubular to annular (ID to OD) was maintained at 3500 psi and held for 15 hours at 220 °F.
13. Post exposure of DPN with low ROC BMG plug in 10.6 ppg brine for 15 hours at 220 °F, test was stopped.

Environment 2

14. Autoclave head was removed from the external pressure vessel and internal pressure vessel was uninstalled.
15. DPN with low ROC BMG plug installed on the inner pressure vessel was removed. Inner pressure vessel and external pressure vessel were cleaned, and chemical waste was disposed. DPN with low ROC BMG plug was inspected, photographed and evidence of reaction, if any recorded. DPN with low ROC BMG plug exposed in 10.6 ppg brine as above was installed back on cleaned internal pressure vessel.
16. Outer pressure vessel was filled with solutions prepared (see above) composed of equal parts (50% brine, 50% breaker) of 10.6 ppg brine and breaker.
17. Breaker-brine solution level in the outer pressure vessel was not exceeding a height of 10" from the top of the autoclave. Breaker-brine solution was heated to 220 °F.
18. The pH value of breaker-brine solution at 220 °F was assessed and recorded.
19. The internal pressure vessel was filled with breaker-brine solution, re-installed on bottom of the head of the external pressure vessel. System was sealed shut and heated back to 220 °F.
20. The inner pressure vessel was pressurized, and pressure differential (ID to OD) was maintained at a minimum of 3500 psi pressure differential for 8 hours or until the nozzle (DPN) breakthrough occurred.

21. On pressure communication, test was stopped. The time for DPN with low ROC BMG plugs to break through was recorded. DPN with low ROC BMG plug was removed, examined, and photographed.
22. The pH value of breaker-brine solution was assessed immediately at conclusion of the test.

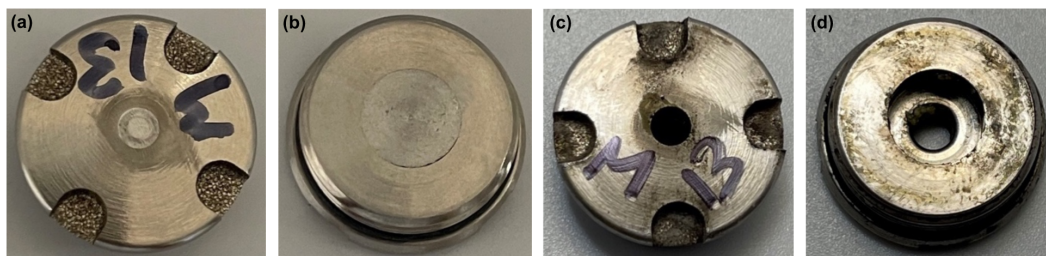


Photo i—(a) Representative DPN with low ROC BMG plug before Test (b) Representative DPNs' with low ROC BMG plug after exposure for 15 hrs. in buffered brine followed by breakthrough (breach) (c) Front (d) Back

Detailed Test Procedure and Equipment

Env. 1 (10.6 ppg brine) followed by Env. 2 (Solution composed of equal parts, 50% brine, 50% breaker of 10.6 ppg brine and breaker).

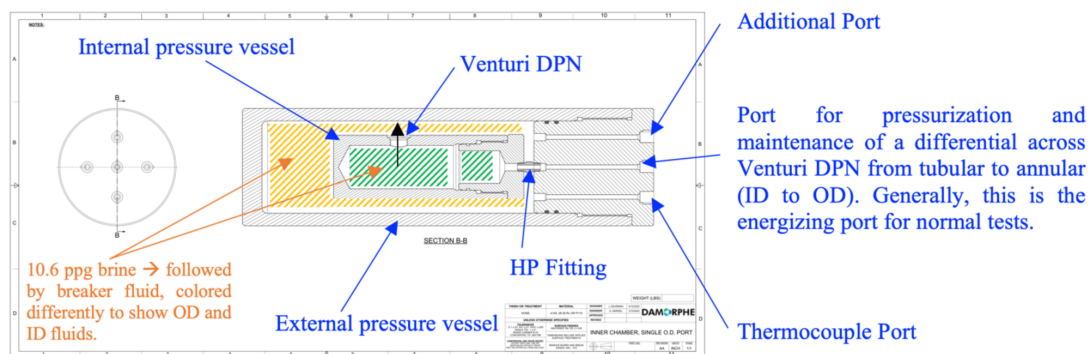


Figure 5—Schematic of specially designed pressure vessels (System): Internal and External pressure vessels to accommodate DPN with low ROC BMG plug with mechanism to expose it to noted weight of fluid, in this case 10.6 ppg brine followed by breaker fluid while being pressurized from tubular to annular that is ID to OD for a set time duration; Assembly allows subsequent reversing, sustaining a differential pressure across DPN, maintaining a specified minimum, to access ability of DPN with low ROC BMG plug to maintain pressure and temperature integrity, reactivity and break-through, if any.



Photo ii—(a) Upper Vessel Heads (b) with Inner Vessels with Nozzles (DPN) Installed (c) Test Set Up for Two Tests (DPN with low ROC plug)

Electron Microscopy and Microstructural Characterization

Dissolvable Plugged Nozzle, a temporary barrier assembly from Salinity Insensitive Metallic-glass Barrier Assembly (SIMBA BL-F) is engineered through novel powder metallurgy route, wherein design encompasses a plug with tailored rates of degradation in client specified fluids.

Substructural examination of bulk HEA partially vitrified nano crystalline alloy samples was conducted using a Philips CM20 TEM operating at 200 kV. Microstructural characterization and novel nano-feature imaging were also conducted in a FEI Tecnai TF-20 FEG/TEM operated at 200 kV in bright-field (BF) TEM mode and High Angle Annular Dark Field (HAADF)-STEM mode. X-ray spectra were acquired on an EDAX system using spot mode and with a nominal STEM probe diameter of about 3 nm.

For TEM samples gauge sections of deformed specimens were cut and mechanically polished down to a mirror finish by final lapping with alumina slurry of 0.1 microns. Once a thickness of less than 70 microns was obtained, approximately 2 mm size discs were punched and affixed to a copper ring using M-Bond epoxy phenolic resin. The gauge sections were then dimpled, and ion milled under liquid nitrogen temperature till the thinned area was electron transparent. Some TEM-ready samples were additionally prepared using the in-situ FIB lift out technique on a FEI Strata 400 Dual Beam FIB/SEM. The sample was capped with FIB e-beam deposited Pt followed by I-beam deposited Pt over the targeted area prior to FIB milling.

Results and Discussions

Results: Operational Testing of Industry-First, BR-DPN with partially vitrified HEA degradable plug

Here we present a "Case Study" where for the very first time, a DPN is designed with a high entropy alloy (HEA) plug, with a partially vitrified nanocomposite degradable matrix. The nozzle is designed to operate under 3,500 psi differential pressure (DP) at 220 °F across the nozzle, is unreactive to oil-based drilling-fluids aka RDFNAF (NAF), retains its mechanical-integrity for up-to 15 hours of exposure in 10.6 ppg (NaBr/NaCl) buffered heavy-brine and a minimum of 8 hours in acidic breaker (a pH of 2 at temperature), subsequently breaking through in less than 4 days.

Between direct brine breakthrough and operational tests encompassing 15-hour exposure to 10.6 ppg brine followed by exposure to 50/50 10.6 ppg brine/breaker under pressure and temperature representing field conditions, over 30 tests were conducted in this V&V round. Selected tests are organized in [Table 3](#).

Based on results of testing SIMBA BL-F qualified to (a) 8% NaCl = 1.05 SG, (b) 21% NaCl + 9% NaBr = 1.25 SG, (c) 1% NaCl + 43% NaBr = 1.49 SG (per client DPN design specifications) for 48 hour holds under 5000 psi differential across the DPN, operational tests with SIMBA BL-F were conducted. Unfortunately, it failed short (~3 hours till breakthrough in inhibited 50/50 10.6 ppg brine/Breaker I after 15 hours of exposure in buffered 10.6 ppg brine) and did not meet field test requirements ([Figure 6](#)). In agreement with our client and partners, to conduct tests with slower blends a suite of tests was planned and subsequently conducted.

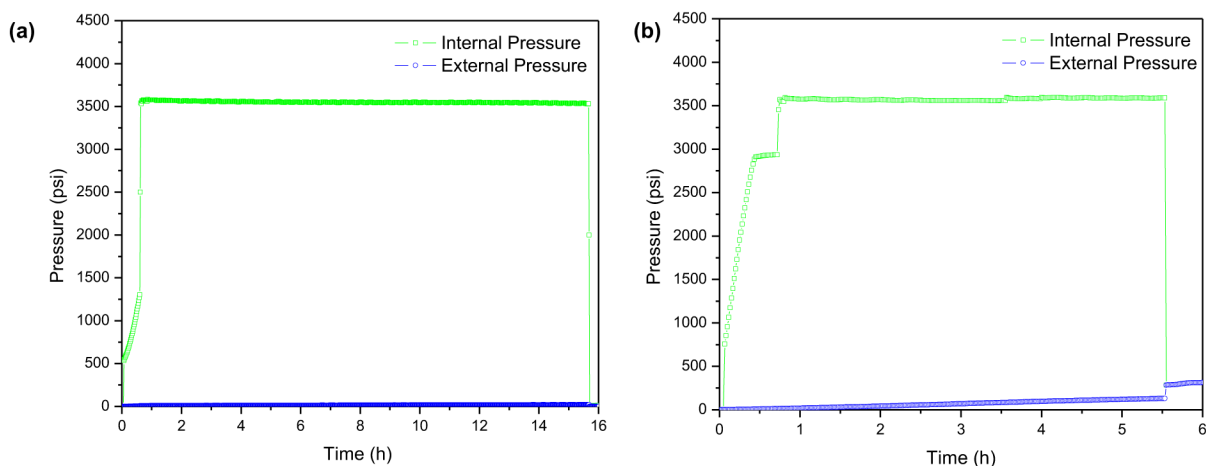


Figure 6—(a) Pressure plot showing internal pressures vessel (green) on which B-F DPN was installed and exposed in low pH 10.6 ppg Brine at 220 °F at 3500+ psi differential pressure (ID to OD) for 15 hours (b) Pressure plot showing internal pressures vessel (green) on which nozzle was installed and exposed in 50-50 Brine-Breaker I at 220 °F at 3500+ psi differential pressure (ID to OD).

Operational tests at 200 °F (lower temperature) with HEA1 clocked 8 ± 1 hr. in inhibited 50/50 10.6 ppg brine/Breaker I after 15 hours of exposure in buffered 10.6 ppg brine at 200 °F. Direct breakthrough tests at 220 °F run in inhibited 10.6 ppg brine clocked 25 ± 1 hr.

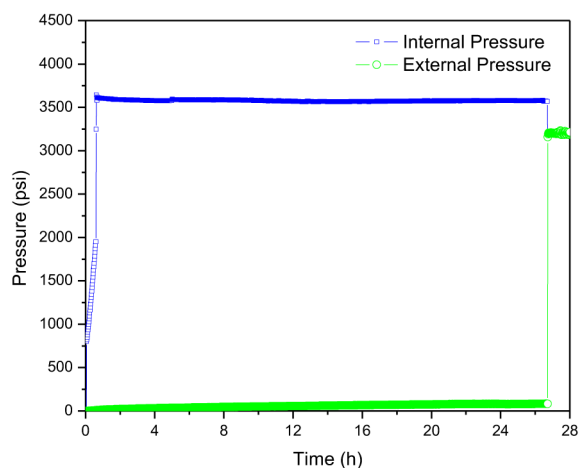


Figure 7—Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA1 was installed and exposed in 10.6 ppg inhibited (old) Brine at 220 °F at 3500+ psi differential pressure (ID to OD) till breakthrough.

Test Breaker II at 220 °F qualified, where we clocked 9.5 hours for a breakthrough time after 15 hours of exposure in buffered 10.6 ppg brine. While this established a promise, it was too close for comfort and did not offer needed factor of safety (FOS). It was hypothesized that modifying HEA1, slowing its degradation may suffice in meeting DPN specifications. HEA1M was as such designed. In parallel multiple additional blends were designed with slower degradation rates [HEA2 to HEA8] to ensure success.

Tests were conducted exposing HEA1M to new brine and breaker received from the Island wells. A hold time of 10.1 hours in 50/50 10.6 ppg brine/Breaker I at 220 °F after a 15-hours exposure in 10.6 ppg brine qualified HEA1M establishing promise for operational success during field test.

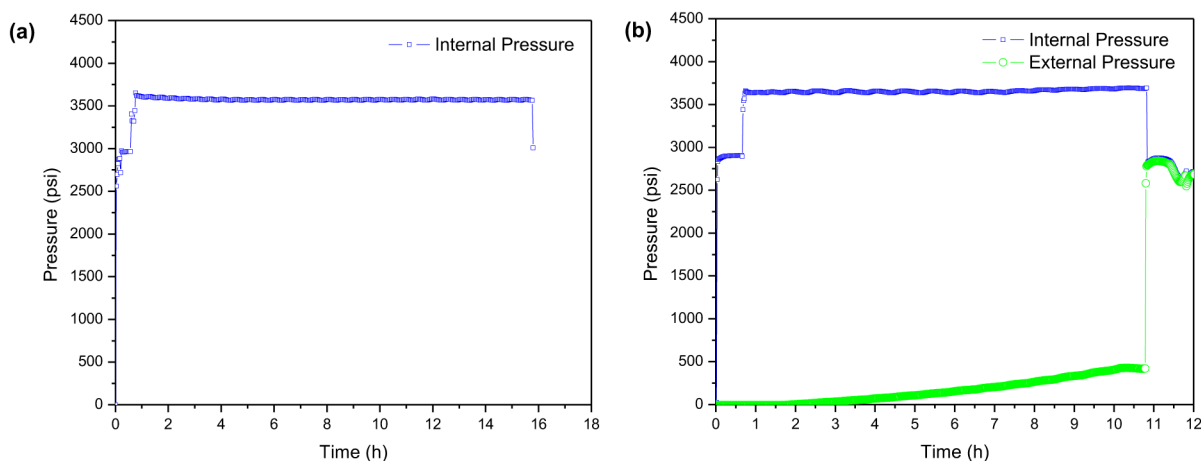


Figure 8—(a) Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA1M was installed and exposed in 10.6 ppg un-inhibited (new) Brine at 220 °F at 3500+ psi differential pressure (ID to OD) for 15 hours. Test 7 (b) Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA1M was installed and exposed in un-inhibited (new) 50-50 Brine/Breaker II at 220 °F at 3500+ psi differential pressure (ID to OD).

We however noticed severe corrosion of low alloy steel (L80) inner pressure vessel, especially after exposure in the Breaker (Photo iii). An insoluble crystalline corrosion debris coated both the external and internal surfaces of the inner L80 pressure vessel. This corrosion debris also deposited on the DPN, passivated and significantly slowed down degradation reaction, as such these long holds of > 29 hours.

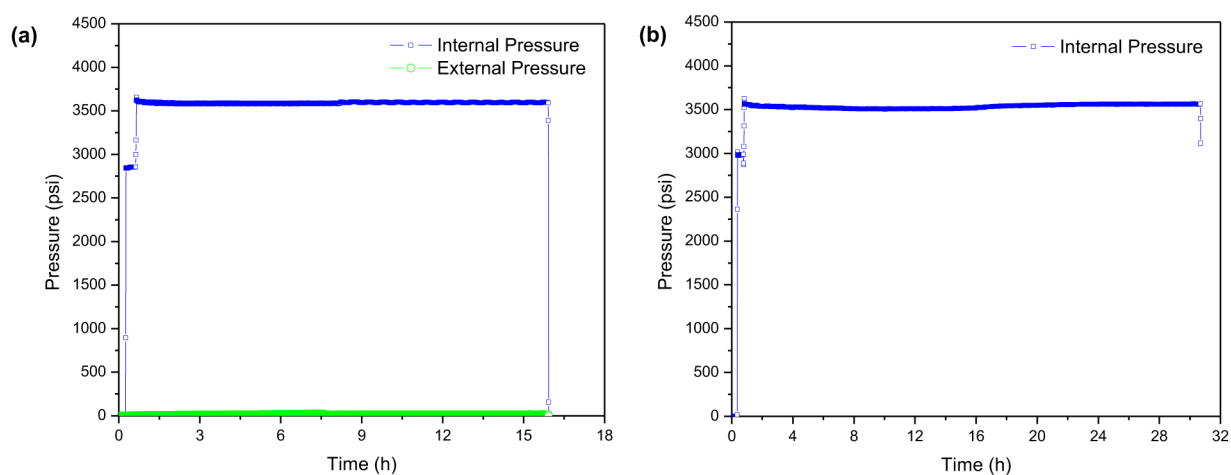


Photo iii—Severe corrosion of the L80 vessel was observed. An insoluble crystalline corrosion debris coated both the external and internal surfaces of the inner L80 pressure vessel. This corrosion debris also deposited on the DPN, passivated and significantly slowed down degradation reaction, as such these long holds of > 29 hours.

The Brønsted-Lowry defines acids as proton (H^+) donors and bases as proton acceptors. Free hydrogen ions don't exist as isolated ions in water. Instead, they immediately combine with water molecules (H_2O) to form hydronium ions (H_3O^+). Water (H_2O) acts as a base in this context, accepting the proton (H^+) from the acid to form hydronium (H_3O^+) ions. The higher the concentration of hydronium ions in a solution, the more acidic the solution is. The pH scale is a measure of this concentration, with lower pH values indicating higher acidity. During exposure, when a metal reacts with an acid, hydronium ions act as oxidizing agents, accepting electrons from the metal to form hydrogen gas and a metal salt (corrosion by-product). This process is a type of redox reaction, where the metal is oxidized, and hydronium ions are reduced. The extent of this reaction depends on the metal's reactivity and the strength of the acid. Depletion of hydronium ions in a solution increases its pH (makes it less acidic, more basic) and renders it less reactive.

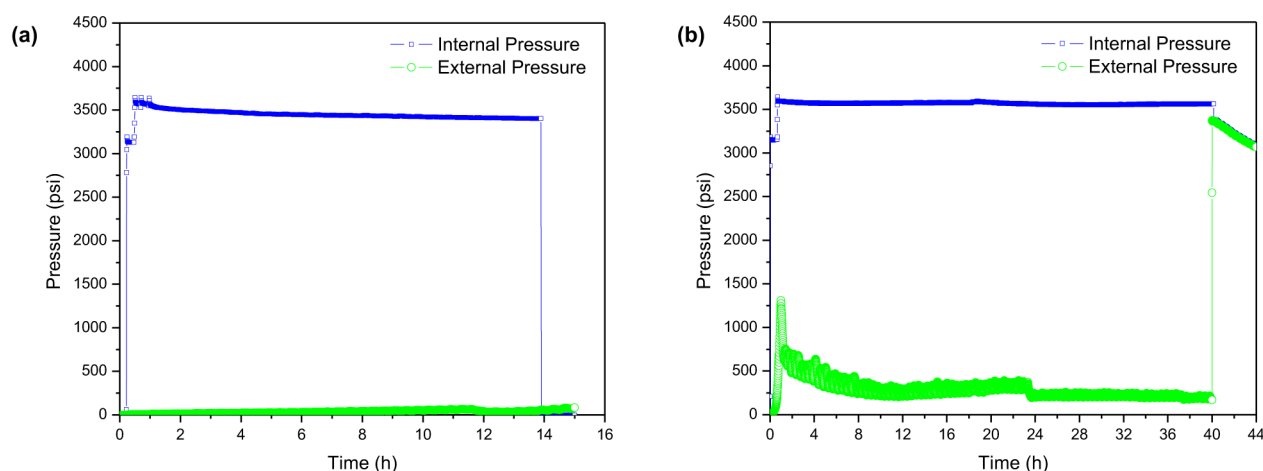
In a L80 vessel (representative of the L80 injection Island well completions), we envisioned that much of the H_3O^+ ions in given volume of Breaker I were consumed during the corrosion of the vessel itself, slowing down the effectiveness of the Breaker I to degrade the DPN, resulting in the longer time needed for a breakthrough.

This theory and caution to test our hypothesis, we decided to expose both HEA1 and HEA1M in the new Brine and Breaker. Operational tests were conducted. There was some corrosion of the L80 internal vessels during exposure to 10.6 ppg brine, however not significant. During Phase 2 exposure in 50/50 10.6 ppg brine/Breaker I, we did not observe a breakthrough in > 29 hours (vs, expected 12 ± 2 hours) (Figure 9 – case HEA1). This led us to stop the tests and investigate. We could not open the pressure vessels even heating it to 500 °F and applying > 1000 ft-lbs. of torque. We were constrained to take it to an external torque machine and subject it to enormous force at elevated temperature to open it, damaging all 3 new L80 pressure vessels build for these tests, rendering them unusable.



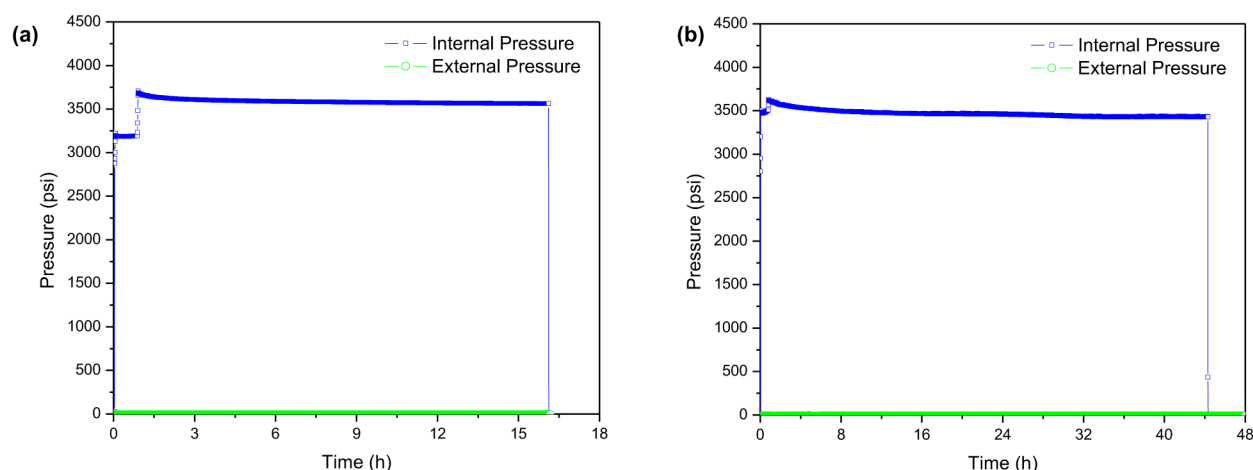
Test 9—Case HEA1: (a) Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA1 was installed and exposed in 10.6 ppg un-inhibited (new) Brine at 220 °F at 3500+ psi differential pressure (ID to OD) for 15 hours. Test 9 (b) Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA1 was installed and exposed in un-inhibited (new) 50-50 Brine/Breaker I at 220 °F at 3500+ psi differential pressure (ID to OD).

General physical appearance of new 10.6 ppg brine and breaker I (vs. former inhibited batches) demonstrated a marked difference in pH, color and viscosity of the fluids and their ensuing behavior (reactivity). To determine the root cause, pH of old vs. new 10.6 ppg brine and breaker I were measured. The new breaker tested to a pH of ~ 2 vs. inhibited former batch tested to a pH of near neutral at STP (pH ~ 6.88). It was discovered that the 10.6 ppg brine and breaker I was devoid of additives (corrosion inhibitors), basically an acid, as such observed corrosion. However, given upcoming well schedules and challenges in getting the fluids from field transferred to conduct complete suite of tests, a solution with the slowest formulation was requested, first assessing repeatable breakthrough in new brine and conducting operational tests in a CRA vessel equivalent to production well completions (13 Cr). On an expedited basis 3#s CRA pressure vessels were manufactured and operational testing initiated. In this case, the 10.6 ppg brine did not affect the vessel metallurgy and resulting activity of H_3O^+ ions in the fluid accelerated the breakthrough of HEA1M DPN in less than 15 hours (Figure 10 (a)).



Test 10—(a) Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA1M was installed and exposed in 10.6 ppg un-inhibited (new) Brine at 220 °F at 3500+ psi differential pressure (ID to OD) till breakthrough. Clocked 13.4 hours to breach. In this case, the brine did not react with the vessel metallurgy at all and resulting activity of H_3O^+ ions in the fluid accelerated the breakthrough of HEA1M DPN in less than 15 hours. **(b)** Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA8 was installed and exposed in 10.6 ppg un-inhibited (new) Brine at 220 °F at 3500+ psi differential pressure (ID to OD).

A slower blend HEA2 was selected and exposed to brine and breaker exposure tests. This fared better, held 15 hours in new 10.6 ppg brine, however clocked 3.3 hours in new 50/50 10.6 ppg brine/Breaker I during operational test. HEA8 a much slower blend was selected to assess boundaries, skipping sequentially slower blends: HEA3 through HEA7. It clocked nearly 40 hours for a direct breakthrough in the new 10.6 ppg brine (Figure 10 (b)). Client requested to test faster blends in case they decide not to run a breaker. Running HEA6 gave us > 43 hours in new 50/50 10.6 ppg brine/Breaker I after a 15-hour exposure in new 10.6 ppg brine (Figure 11). This established promise that HEA5 may offer the right degradation rate to ensure field tests success.



Test 11—(a) Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA6 was installed and exposed in 10.6 ppg un-inhibited (new) Brine at 220 °F at 3500+ psi differential pressure (ID to OD) for 15 hours. **(b)** Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA6 was installed and exposed in un-inhibited (new) 50-50 Brine/Breaker I at 220 °F at 3500+ psi differential pressure (ID to OD).

Given fluid constraints and DPN availability, 3 tests with HEA5 were conducted to establish repeatable and reliable breakthrough in 10.6 ppg brine. We achieved 50 ± 10 hours for a direct breakthrough (Figure 12(a) and (b)). Subsequently we conducted operational tests in old 10.6 ppg brine, and a combination of 50/50 old 10.6 ppg brine/old Breaker I and 50/50 old 10.6 ppg brine/new Breaker I. Operational tests in the

new un-inhibited 50/50 10.6 ppg brine/Breaker I bought us >37 hours till breakthrough after 15 hours of exposure in uninhibited 10.6 ppg brine (Figure 13 (a) and (b)).

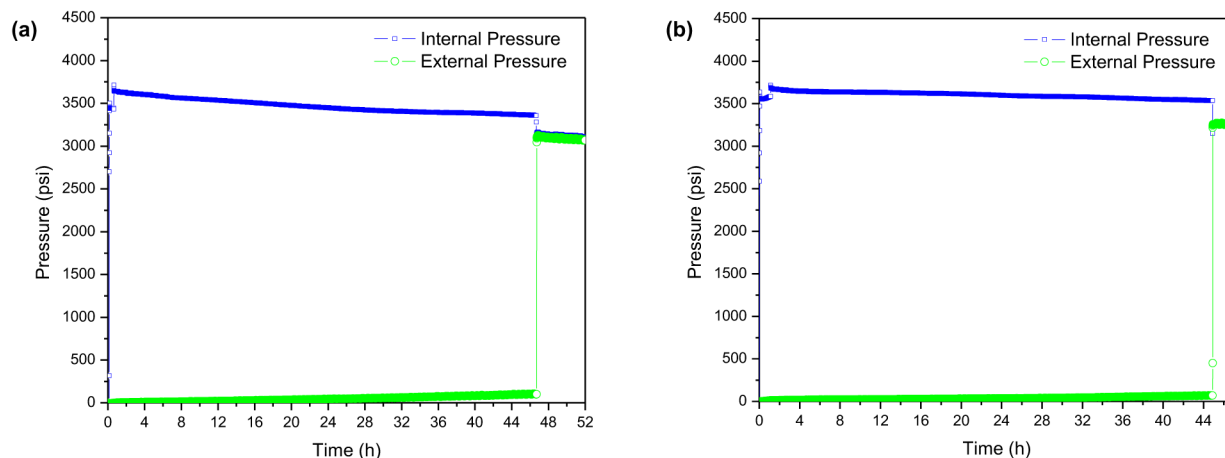


Figure 12—Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA5 was installed and exposed in 10.6 ppg un-inhibited (new) Brine at 220 °F at 3500+ psi differential pressure (ID to OD). Clocks 46.1 hours to breach. (b) Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA5 was installed and exposed in 10.6 ppg un-inhibited (new) Brine I at 220 °F at 3500+ psi differential pressure (ID to OD).

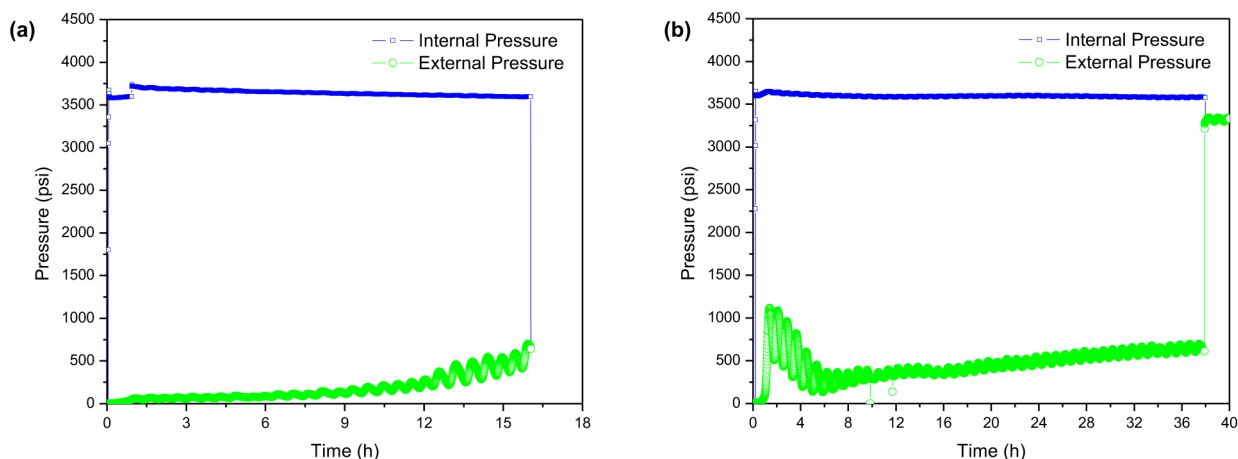


Figure 13—(a) Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA5 was installed and exposed in 10.6 ppg inhibited (old) Brine at 220 °F at 3500+ psi differential pressure (ID to OD) for 15 hours (b) Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA5 was installed and exposed in un-inhibited (new) 50-50 Brine/Breaker I at 220 °F at 3500+ psi differential pressure (ID to OD).

HEA5 plugged DPNs' were also exposed to former batch of inhibited Breaker I from 2024 Phase – I test program. Our HEA5 blend lasted 15+ hours in the old 50/50 10.6 ppg brine/Breaker I after 15 hours of exposure in the old buffered 10.6 ppg brine (Figure 14). The rationale for HEA5 to maintain mechanical integrity (>37 hrs. vs. reduced time of 15+ hrs.) is likely (1) due to breakdown of the inhibited Breaker I over time (2) Because of the added inhibitors it might have coated pressure vessel surfaces, preventing reaction between the fluid and vessel metallurgy, preserving activity of the fluid. The Breaker I with hydronium ions (H_3O^+) preserved may have resulted in accelerated degradation of the installed DPN. This is representative and the most likely case, wherein an active, flowing inhibited breaker will interact with DPNs' in the wellbore.

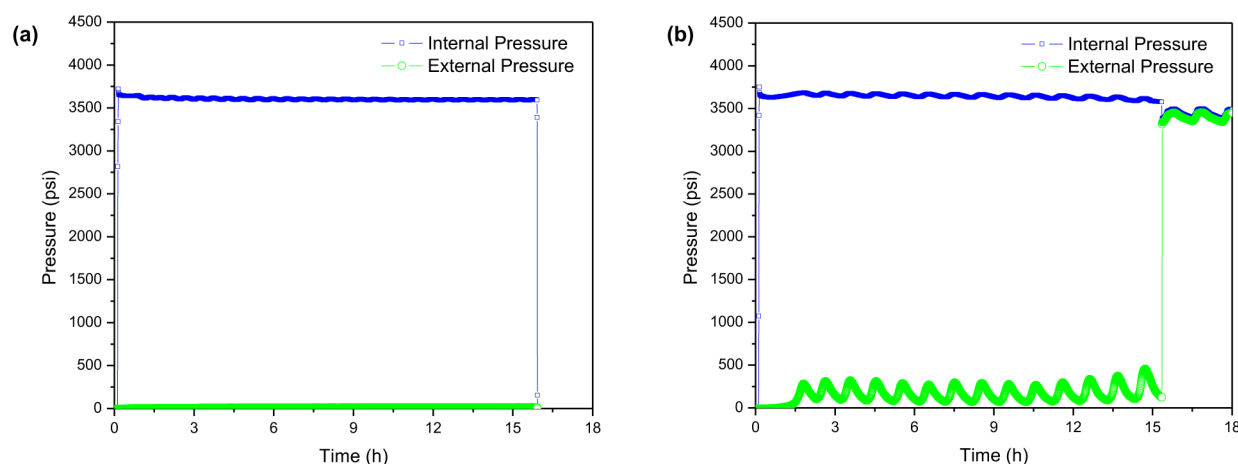


Figure 14—(a) Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA5 was installed and exposed in 10.6 ppg inhibited (old) Brine at 220 °F at 3500+ psi differential pressure (ID to OD) for 15 hours. (b) Pressure plot showing internal pressures vessel (blue) on which DPN with low ROC BMG plug, HEA5 was installed and exposed in inhibited (old) 50-50 Brine/Breaker I at 220 °F at 3500+ psi differential pressure (ID to OD).

HEA5 (and all developed formulations) has repeatedly survived 15 hrs. in 10.6 ppg brine and > 15 hours (wherein 8+ hrs. is the qualification metric) in uninhibited breaker or its old, inhibited counterpart. Based on test data we have good confidence in running HEA5 successfully in the upcoming well as long as the brine and breaker fluids deployed (run in hole (RIH)) are not very chemically divergent from the one supplied to us.

Discussions

In the past, tests to evaluate the ability of a SIMBA BL-F to hold pressure and mechanical integrity when aged (exposed to) in RDFNAF at an elevated temperature of 220 °F subjected to a tubular to annular differential pressure of 3500 psi for 20-days. The successful conclusion of the test established inertness of our high entropy alloy partially (or fully) vitrified core forming a gas tight seal to NAF and its superiority compared to other commercial offerings. This challenging test was requested as a pre-requisite for field introduction of SIMBA DPNs'. Subsequently, both pressure and mechanical integrity of the SIMBA DP Nozzles have been established during exposure to a variety of brines. SIMBA BL-F has been qualified to 5000 psi for up to a 48-hour hold in 10.6 ppg and a variety of different brines. After comprehensive tests exposing different blends in both inhibited and un-inhibited fluids, given success criteria for DPNs' to reliably maintain mechanical integrity for 15 hours in inhibited and buffered brine followed by 8-hours (+FOS) or more in 50/50 10.6 ppg brine/Breaker I, from repeat test data, HEA5 with apposite degradation rate window stood out as THE promising candidate.

Vitrification and Machining Challenges

Machining bulk metallic glasses (BMGs) presents unique challenges due to the brittle nature of the alloy and their tendency to deform unevenly, often exhibiting serrated chip formation due to shear localization. Cutting speed and depth of cut influence temperature during machining. While thermal softening can aid machining, low thermal conductivity of BMGs can lead to localized heating and affect surface quality. Achieving a high-quality surface finish is a major challenge, and factors like cutting parameters and tool material selection play a crucial role. Advancements in both traditional and non-traditional machining methods are being explored to overcome these issues. Key challenges include achieving precise geometries, maintaining surface integrity, and managing chip formation. Diamond turning, milling, micro-machining, Electrical Discharge Machining (EDM), Laser are being investigated.

Processing of HEA5 material was conducted at elevated temperatures and pressures as such it was necessary to maintain the temperature for the vitrification die above the materials glass transition temperature (T_g) (Figure 15). Careful approach to machine the vitrified BR-DPNs' to final form has been adopted.

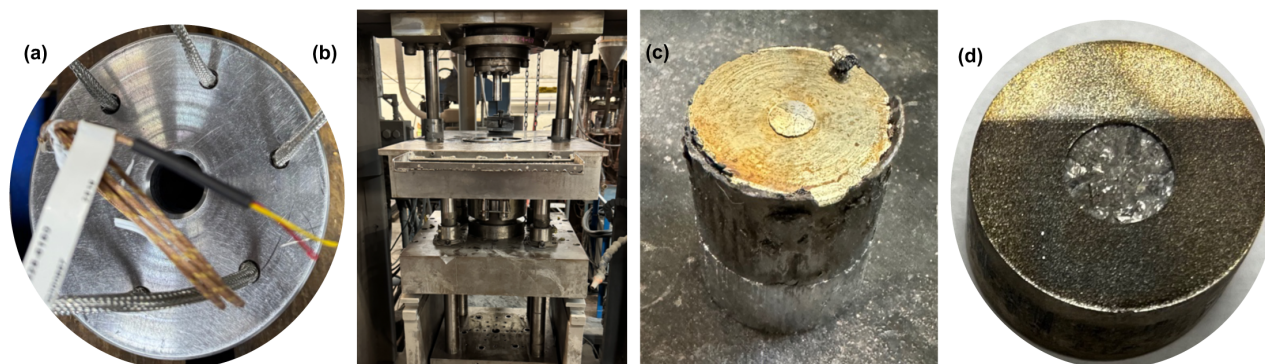


Figure 15—(a) Heater elements connected to vitrification die to pot HEA preforms into DPN blank. Processing is conducted at elevated temperatures and pressures as such it is necessary to maintain the temperature for the vitrification die above the materials glass transition temperature (T_g) (b) Photograph of vitrification press (c) Preform vitrified in die (d) shear facets evident in the partially vitrified alloy, promoted during machining of the brittle BMG.

Quality Control: X-Ray

X-ray imaging or digital radiography is a non-destructive testing method used to detect cracks and other internal flaws in metal components. Different materials, and variations in density within the same material, absorb X-rays at different rates. Cracks and voids, being less dense than solid metal, will allow more X-rays to pass through, wherein areas with more X-ray penetration appear darker and areas with less penetration appear lighter. Cracks and other flaws are revealed as darker areas or variations in the image pattern, allowing identification of their location and size.

This methodology has been adopted to ensure quality and mechanical integrity of the BR-DPN, wherein the acceptance criteria is observance of no flaws or evidence of cracks or debonding in the filled core of the BR-DPN.

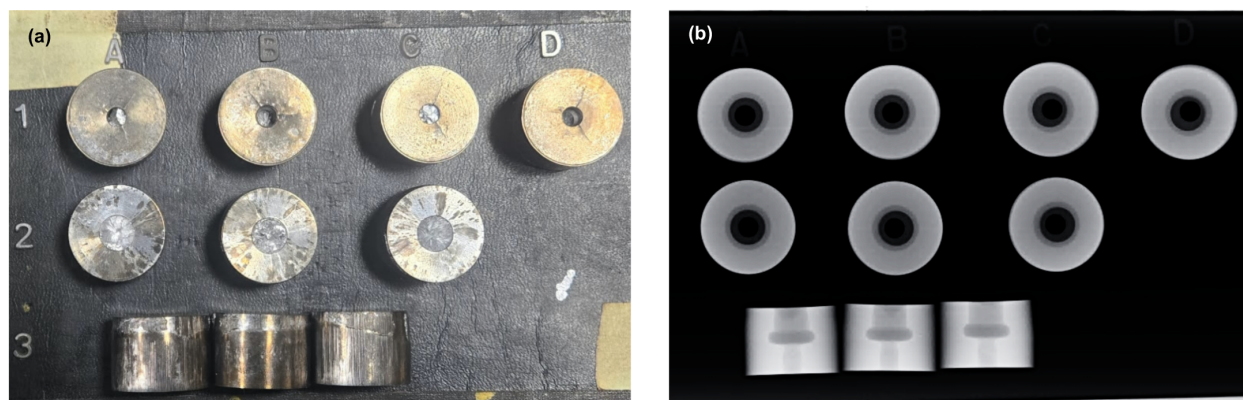


Figure 16—(a) Actual potted BR-DPN blanks arranged to obtain X-Ray images (b) Plan and side view X-Ray images of BR-DPN blanks potted with vitrified HEA alloy. No cracks are evident in the fully filled core of the BR-DPN, a measure adopted to ensure quality and mechanical integrity of the BR-DPN.

Concluding Remarks and Innovative Concepts

Fast-track engineering to expeditiously deploy new technologies in upstream oil and gas heavily relies on timely agreement and synergy between the engineering team and stakeholder on (1) tool architecture, thus

specifications (2) qualification criteria(s) (3) operational envelope and RIH environment (4) verification & validation strategy, among others. Strategic partnership between Academia, Engineering and Service, abetted by streamlined flow of coherent information regarding success criteria for an effective tool design.

A case of accelerated product development through Nippon Foundation-Deepstar Collaborative Technology Development Grant Program high-efficiency "Breaker Resist" plugged nozzles have been jointly developed with Daido Steel of Japan and Baker Hughes as commercialization partner. Key novelties of this industry-first design are (a) Not having to pick-up a dedicated displacement work-string, thus significant time-savings of up to 5-days (b) Increase probability of running liner to TD through the ability to wash to bottom (c) Less fluid losses to the reservoir formation (d) Effective wellbore cleaning, among others (e) Ability to deploy tracers, field- tested and proven to identify breakthrough of BR-DPNs' (f) Unparalleled economics.

A direct breakthrough time of 50 ± 10 hours for HEA5 was achieved in new uninhibited brine. Subsequently operational tests in old 10.6 ppg brine, and a combination of 50/50 old 10.6 ppg brine/old Breaker I and 50/50 old 10.6 ppg brine/new Breaker I bought us between 15 to >37 hours till breakthrough. Our HEA5 (and all developed formulations) has repeatedly survived 15 hrs. in 10.6 ppg brine and > 15 hours (wherein 8+ hrs. are the qualification metric) in 50/50 10.6 ppg brine/uninhibited Breaker I or its old, inhibited counterpart. Based on test data we have good confidence in running HEA5 successfully in the upcoming well as long as the brine and breaker fluids deployed (run in hole (RIH)) are not very chemically divergent from the one tested in.

As a closing note, the game changing brine insensitive ultra-high strength, thermally stable, dissolvable nano-composite alloy behind SIMBA BR-DPN's plug, can be easily tailored, making it a unique offering in this technology space. Recommended shelf life is 6-months (to a year at most). For the field trials a plug will be designed (engineered to order) to survive needed hours in breaker fluids post exposure in RDFNAF, brine(s) of a given specific gravity.

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